Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 411–421

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

CNN Fusion: A Promising Technique for Ophthalmic Disorder Diagnosis

Ankur Biswas[a]*, Rita Banik[b]

[a]*Tripura Institute of Technology, Narsingarh, West Tripura, 799009, India*
[b]ICFAI University Tripura, Kamalghat, West Tripura, 799210, India
abiswas.tit@gmail.com

Abstract

Ophthalmic disorders represent a major global health problem leading to visual impairment and even blindness if not detected and treated early. Deep learning is a popular approach where pre-trained models are now often used to diagnose a variety of diseases from different medical images. In order to provide an accurate and reliable categorization of ophthalmic illnesses, this study suggests a fusion or ensemble technique that harnesses the power of pre-trained models like ResNet50, DenseNet and EfficientNet. The proposed ensemble model enhances the performance by incorporating the benefits of three architectural paradigms, guiding the convergence of each one's distinct strengths to achieve improved overall classification result. The ensemble architecture has undertaken inclusive validation on multiple fundus image datasets, showing improved performance over lone models. An accuracy of 92% demonstrates the potential of ensemble model in the categorization of eye diseases like cataract, diabetic retinopathy, glaucoma and normal eyes. In addition to achieving cutting-edge accuracy, the ensemble also offers an AUC-ROC score of 1.00. This technique presents a viable alternative for boosting the accuracy and reliability in identifying various eye illnesses by combining diverse model capabilities. The empirical data and conclusions show that the ensemble model has the potential to make a substantial contribution to the field of ocular medical imaging and patient care by enabling precise and timely treatments that may significantly improve the lives of millions of people worldwide who are impacted by ophthalmic diseases.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Ophthalmic disorder; Deep learning; Classification; Pre-trained model; Ensemble

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.231

1. Introduction

Ophthalmic disorders, which can range from mild problems to serious abnormalities, pose a significant threat to world health and have comprehensive effects. These conditions need fast and accurate diagnoses for efficient management and treatment since they might impair vision and perhaps result in blindness [1-3]. For diagnosis and therapy, it is essential to classify ophthalmic illnesses accurately. However, manual inspection and categorization of eye problems takes a lot of time and requires specialized knowledge. The complex structure and function of the human eye act as a window into the overall state of the body. The development of digital imaging has made it possible to take non-invasive, high-resolution fundus photographs, giving physicians a new level of clarity when examining ocular tissues and spotting anomalies [4-5]. The rapid progress in deep learning has significantly changed medical image analysis, especially in the field of ophthalmology. These systems can leverage large-scale training data and learn complex patterns in medical images, leading to improved accuracy and efficiency. Due to the complicated patterns associated with illnesses and the intricacy of the anatomical systems of the human eye, identifying and diagnosing eye disorders from fundus images presents significant problems. In order to improve diagnosis accuracy, this study offers a novel technique that uses deep learning models. Deep learning algorithms have showed a lot of potential in diagnosing eye diseases, especially diabetic retinopathy and glaucoma [6-10]. Here are a few examples of how deep learning has been utilized for this purpose: To categorize distinct types of ocular disorder, deep convolutional neural network (CNN) algorithms were combined with clinical scaling systems. These systems can identify and categorize retinal disorders including neovascular leakage, as well as give a full evaluation of common imaging modalities. Based on a hybrid deep learning method, a computer-aided diagnosis system for recognizing various diabetic retinopathy grades has been created. This system uses deep CNN architectures to identify normal and abnormal retinal images and gives a critical evaluation of deep learning approach performance. Deep learning-based evaluation of fundus images has been demonstrated to be beneficial in the automation of large-scale glaucoma detection and screening programmes. Almost every eye condition may be identified, localised, and quantified using machine learning and deep learning technologies. A deep learning method has been created to detect neovascular leakage in diabetic retinopathy images using ultra-widefield fluorescein angiography. This technique is capable of detecting neovascular leakage, which is a key hallmark of diabetic retinopathy. Pre-trained models are neural network models that have been trained on big datasets and may be used to begin a variety of deep learning tasks like image classification viz., eye disorders classification.

The above applications incited that the accuracy, effectiveness, accessibility, and scalability of eye disease identification might all be improved with deep learning-based methods, which would enhance patient outcomes and save healthcare costs. Hence, this study intends to propose a unique technique that makes use of the advantages of many pre-trained deep learning models, each of which is distinct by its architecture and feature extraction capabilities. The fusion of these models is hoped to overcome the shortcomings of individual models and provide a more reliable, and thorough approach to the categorization of eye diseases like cataract, diabetic retinopathy, glaucoma and normal eyes. Combining several model capabilities helps to reduce the biases and uncertainties that occur with using a single method, which eventually leads to more accurate diagnostic results [11-12]. A number of well-known CNN architectures that were already trained on the ImageNet dataset are enhanced to perform multi-class classification of eye diseases. These include ResNet50, DenseNet and EfficientNet [13-15]. The final forecast is then created by combining the outputs from three models using a fusion or ensemble technique. On benchmarking Indian Diabetic Retinopathy Image Dataset (IDRiD) and EyePACS public datasets for eye diseases, the proposed ensemble model [16-17] is assessed. Our findings demonstrate that utilizing a group of pre-trained models instead of just one CNN model resulted in higher classification performance. This implies that incorporating data from several model designs might increase the precision of classifying eye diseases. Finally, the ensemble model's accuracy of 92% represents a substantial advancement in machine learning and predictive modelling. This achievement demonstrates the capability of leveraging the advantages of several pre-trained architectures to produce a more reliable and precise prediction system. Moreover, the model achieved an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) score of 1.00 in categorizing eye disorders.

The theoretical foundations of deep learning in medical imaging will be explored in more detail in the next sections of this study, revealing the basis upon which our fusion technique is built. The methodology itself will be described, including model selection, fine-tuning, and forming the ensemble. The empirical results that demonstrate the merits

of the proposed strategy through careful assessment and comparison will be explained. The implications of the findings for clinical practice and the bigger picture of medical image analysis for future implication will also be examined in the final section.

2. Background

Significant efforts have been focused towards the enhancement of eye disease detection using diverse approaches in the area of related studies. Deep learning, namely the use of convolutional neural networks (CNNs), has emerged as a popular method for accurate disease categorization. Several studies have shown that CNN designs can achieve good diagnosis accuracy across a wide spectrum of eye diseases. Furthermore, transfer learning algorithms that leverage pre-trained models from other domains have showed promise in improving disease detection model performance. While individual models have shown noticeable results however, certain models that are now in use might not be able to adapt well to different datasets or imaging situations. If this restriction is used on fresh or untested data, performance can suffer. Moreover, the capacity of models to effectively diagnose less common eye disorders may be impacted by bias towards the majority class in datasets that were used to train them. Additionally, eye disorders can present in a variety of ways, and current models may find it difficult to account for the wide range of visual cues connected to specific conditions. Therefore, the current research takes a fresh approach by suggesting a fusion strategy that capitalizes on the combined capabilities of numerous models This method not only reduces individual model biases but also helps to improve diagnostic accuracy and resilience. Pre-trained models have demonstrated their mastery of feature extraction and knowledge transfer, frequently arising from architectural advancements like ResNet, DenseNet, EfficientNet etc. [18-22]. By applying these models to medical images, illness detection and early intervention techniques might be revolutionized. The current study builds on prior research in order to provide a completer and more reliable diagnostic paradigm for eye illnesses.

2.1. ResNet50 architecture

A convolutional neural network with 50 layers is called ResNet-50. It is a well-liked deep learning model applied to applications involving picture categorization. Five stages make up the ResNet-50 model, each of which has an identity block and a convolutional block. There are three convolutional layers in each identity block and three convolutional layers in each convolutional block. Over 23 million trainable parameters exist in the model. There are six components to the ResNet-50 architecture: Input preparation, CFG blocks and fully connected layer

The ResNet-50 model makes advantage of skip connections, which let data skip one or more network levels and go straight to a subsequent layer. This makes it possible to build deeper neural networks and helps to solve the vanishing gradient problem. It employs residual learning, which involves adding shortcut connections to each pair of 3X3 convolution layers. These contribute to addressing the degradation issue, as deeper networks are more difficult to train. There are around 25 million parameters in the network. It accepts 224x224x3 pixel inputs because it was created for the ImageNet dataset. The ResNet50 model is supplied as a pre-trained model that can be used right away for new dataset features or fine-tuning.

2.2. DenseNet-121 architecture

DenseNet-121 is a convolutional neural network design from the DenseNet model family. DenseNet-121 was developed primarily for image classification tasks and has been widely utilized in a variety of computer vision applications. Here are some important facts regarding DenseNet-121:

DenseNet-121 has dense connections between layers, which means that each layer is feed-forwardly coupled to every other layer. It has around 8 million parameters, which is much fewer than other networks with comparable performance, such as ResNet. This increases its memory and storage efficiency. This extensive connection improves information flow and gradient propagation across the network. The network is made up of four dense blocks that are linked together by transition layers. The first dense block contains 64 layers, the second and third have 128 layers apiece, and the last one has 256 layers. This extensive connection encourages the reuse of features and allows the network to learn additional discriminative characteristics. To boost computing performance, it adds bottleneck layers.

A bottleneck layer is made up of a 1X1 convolutional layer and a 3X3 convolutional layer. The 1X1 convolutional layer minimizes the amount of input feature maps, which lowers the computational cost, whereas the 3X3 convolutional layer captures more complicated patterns. The Transition layers between dense blocks are included in DenseNet-121. These transition layers minimize the amount of feature maps by downsampling the spatial dimensions of the feature maps. This aids in controlling the expansion of feature maps and managing the network's computational complexity. It employs global average pooling at the network's end to combine the feature mappings into a fixed-length vector. This vector is then used to classify data in a fully connected layer.

2.3. EfficientNetB3 architecture

A convolutional neural network architecture called EfficientNet-B3 was created by Google researchers. It was created to perform image categorization tasks with a decent mix of efficiency and accuracy. Here are some main features of EfficientNetB3:

EfficientNetB3 is a deep neural network design made up of convolutional layer blocks. The architecture is based on a compound scaling system that continually grows network width, depth, and resolution. This allows for more efficient use of computing power while offering higher accuracy. It employs the "MBConv" block, that refers to Mobile Inverted Residual Bottleneck Convolution. This block's goal is to be computationally lightweight while yet retaining a high level of precision. A depth-wise convolution, a point-by-point convolution, and a skip connection comprise it. EfficientNetB3 has demonstrated outstanding scores on a range of image classification targets, including ImageNet, CIFAR-10, and CIFAR-100. It has additionally been shown that its implementation can be more efficient in terms of computation than comparable designs such as ResNet and MobileNet. Deep learning platforms like as TensorFlow or PyTorch may be employed to deploy all the ResNet-50, DenseNet-121, EfficientNetB3 structures. The two platforms offer these models that have been pre-trained and may be fine-tuned for particular purposes or used as feature extraction tools for transfer learning.

3. Methodology

The proposed approach centers on building an ensemble of pre-trained models that have been carefully customized for categorizing eye disorders. Using a wide range of fundus images, each unique model is precisely adjusted. The ensemble forecast is then created by combining the results from each model using a simple voting method. The main goal is to use the combined power of several models to overcome the shortcomings of individual models, thus increasing classification accuracy overall.

3.1. Data collection and Preprocessing

An extensive dataset with a variety of fundus images capturing different eye disorders forms the basis of the study to guarantee robustness and representativeness. The dataset is divided into subsets for training, validation, and testing, enabling objective model evaluation. The dataset of over 4000 retinal fundus photos has been curated from IDRiD, EyePACs dataset, Ocular Disease Intelligent Recognition (ODIR) dataset, High-Resolution Fundus (HRF) Image Database [23-24]. Images of diabetic retinopathy, glaucoma, cataracts, and normal were among the four eye conditions depicted. Each photograph has the proper illness labelled on it by ophthalmologists. To make consistent with the input size of the pre-trained models, photos were preprocessed by making them grayscale and shrinking them to 224x224 pixels. Fig. 1 displays four different types of eye disorders present in the dataset.

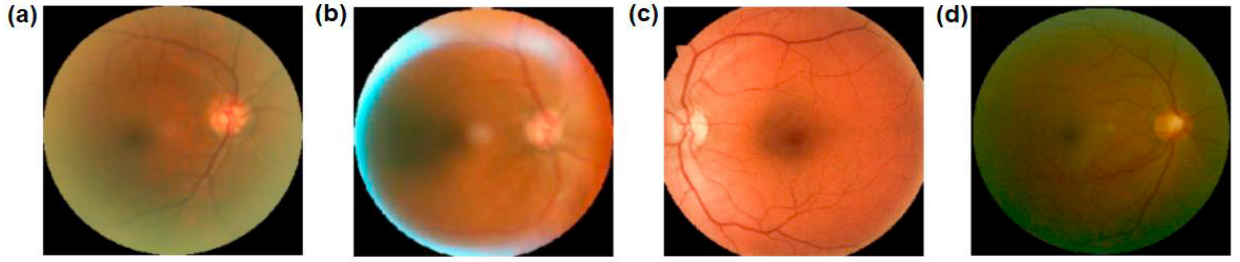


Fig. 1. (a) normal eyes; (b) cataract; (c) diabetic retinopathy; (d) glaucoma.

3.2. Model Architecture

The ensemble model's design effectively integrates various pre-trained frameworks, most notably ResNet, DenseNet and EfficientNet. Each model's convolutional layers work as feature extractors, collecting various hierarchies of characteristics from input fundus images. The retrieved characteristics are subsequently concatenated and provided to fully connected layers that are in charge of final categorization.

Three pre-trained CNN models, DenseNet161, ResNet50, and EfficientNetB3, were combined. These models have successfully served as feature extractors for transfer learning to new domains and applications since they were initially trained on the ImageNet dataset for image classification tasks. Each model's top classification layers were eliminated, and only the convolutional base's pretrained weights were used. For categorizing the four class of eye diseases, new classification heads with softmax activations were introduced.

3.3. Model Fusion

The convolutional bases of each model were retrained and fine-tuned using transfer learning for the purpose of classifying eye diseases. With a learning rate of 0.001, stochastic gradient descent was used to train the models from beginning to finish. The evaluation metric was accuracy, and the loss function was categorical cross-entropy. The models were trained using batches of 32 photos across 10 epochs. To lessen overfitting, data augmentation methods such as random cropping, flipping, and brightness alteration were applied. The final ensemble model predictions were made by averaging the softmax probability predicted by each of the three refined models. A held-out test dataset was used to assess this ensemble model's performance in terms of categorization. The predicted probabilities of three models are $p_1,\ p_2\ and\ p_3$ respectively. A weighted average can be used to combine the predictions, where the weights are proportional to the accuracy of each model:

$$F_{ensemble} = w_1 p_1 + w_2 p_2 + w_3 p_3 \tag{1}$$

where, $w_1, w_2\ and\ w_3$ are assigned weights to models respectively. The weights can be determined using many techniques, such as Bayesian model averaging, or depending on each model's performance on a validation set. Another typical method of combining forecasts is through voting, in which each model makes a prediction and the result is determined by the model that received the majority of votes. When doing classification tasks, where the outcome is a discrete label, this method may be helpful. A straightforward method is followed to combine predictions for classification problems through majority vote:

- Each pre-trained model makes a prediction for every input.
- The class that obtains the most model votes is the final forecast.

$$C_{ensemble} = argmax_c = \sum_1^3 [c_i = c] \tag{2}$$

where c_i represents the classes by the pre-trained models and $[c_i = c]$, the indicator function. When c_i equals c, it returns 1, otherwise it returns 0. The class that maximises the sum of the indicator functions is $C_{ensemble}$, and it

represents the final forecast. Combining predictions from numerous models is a type of model averaging. Predictions can be averaged to assist reduce the influence of individual model biases and errors, resulting in more accurate and consistent forecasts. The chance of one model producing an incorrect forecast is reduced by the availability of other models that can make accurate predictions when there are numerous models making predictions. This strategy may minimize overfitting and increase the combined model's capacity for generalization. Utilizing the complementing data gathered by each model, the forecasts may be combined to gain a more thorough knowledge of the data and perhaps even more accurate predictions.

4. Experimental Results

The discussion explores the relevance of using an ensemble learning technique to classify eye diseases. The integration of diverse models is emphasized in order to develop a comprehensive and more accurate diagnostic framework, hence improving illness detection reliability. The model is thoroughly evaluated, with comparisons to standalone pre-trained models and other classification strategies. The ensemble model's superiority is demonstrated by performance indicators such as accuracy, precision, recall, and F1-score [25-27]. Accuracy is the measure of correct predictions made by a model over all kinds of predictions. Precision is the measure of the correctly predicted positive cases out of all predicted positive cases. Recall is the measure of the correctly predicted positive cases out of all actual positive cases. F1-score is the harmonic mean of precision and recall, providing a balance between the two metrics. The experimental results highlight the model's robustness, generalizability, and potential to affect clinical decision-making. Table 1 presents the precision of each class along with overall accuracy for three stand-alone models

Table 1. Precision and accuracy of pre-trained models.

Disorder class	ResNet50	DenseNet121	EfficientNetB3
cataract	0.89	0.94	0.84
diabetic_retinopathy	0.98	0.86	0.95
glaucoma	0.89	0.70	0.84
normal eyes	0.92	0.99	0.81
Overall accuracy	0.90	0.86	0.87

DenseNet121 had the highest precision of 0.99 when it came to categorizing normal eyes. ResNet50 had the greatest precision of 0.98 when it came to identifying diabetic retinopathy. DenseNet121 and ResNet50 classified cataracts well, with precisions of 0.94 and 0.89, respectively. The glaucoma classification precisions varied from 0.70 to 0.89. In terms of overall accuracy, ResNet50 fared best with 0.90 accuracy, followed by EfficientNetB3 at 0.87 and DenseNet121 at 0.86. Each of the model displayed strengths in diagnosing certain types of eye illnesses, suggesting that an ensemble approach may allow us to capitalise on the complementing performance of the various architectures. When no single model outperforms the others, forming an ensemble is a frequent method. In this situation, when the precision and accuracy scores of ResNet50, DenseNet121, and EfficientNetB3 are quite close, integrating them into an ensemble model is the next step that might enhance classification performance for eye illness diagnosis. The accuracy and loss graph after forming the ensemble is shown in Fig. 2.

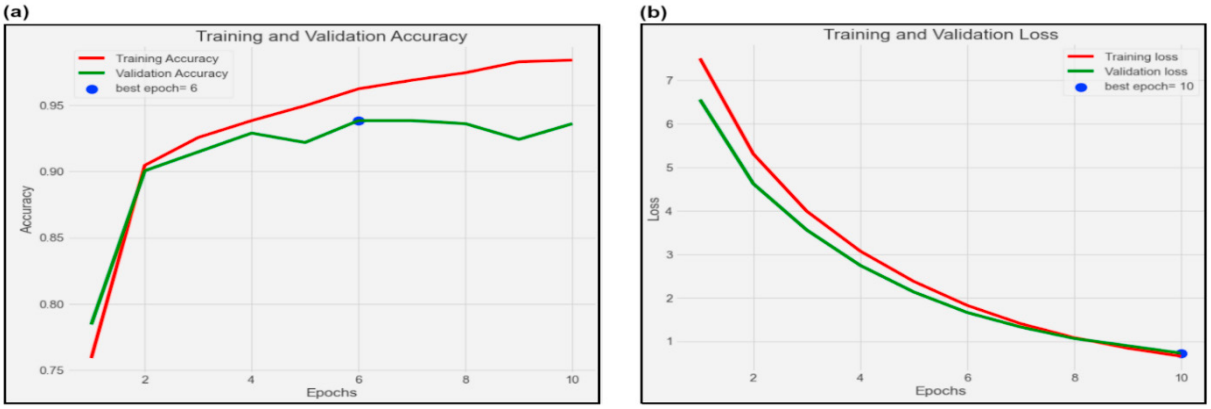


Fig. 2. (a) Accuracy; (b) loss per epoch for the proposed ensemble model

The model is initially tested for 10 epochs. The accuracy and loss graphs visually represent the insights into the models' performance and training development which revels that the accuracy increases and loss decreases as the training proceeds. The accumulation of epochs demonstrates that the best accuracy is attained at epoch 6, indicating the model's capacity to generate precise predictions early in training. Conversely, the loss function converges around epoch 10, confirming the model's capacity to minimize the difference between predicted and actual values. This intricate interaction between accuracy and loss emphasizes the dynamic nature of the training process and highlights the significance of analyzing these measures over epochs to acquire a complete picture of the model's growth. The overall performance of the model is presented in Table 2.

Table 2. Classification report of the proposed ensemble

Disorder class	Precision	Recall	F1-score
cataract	0.97	0.89	0.93
diabetic_retinopathy	1.00	1.00	1.00
glaucoma	0.83	0.95	0.88
normal eyes	0.91	0.85	0.88
average	0.93	0.92	0.92

The model's capabilities lay in precisely detecting occurrences of diabetic retinopathy and exhibiting high performance for cataract identification, with an overall accuracy of 92%. The model correctly recognizes all cases of diabetic retinopathy. Also, the model obtains 97% accuracy, 89% recall, and an F1-score of 93% for the "Cataract" class. This shows that the approach is effective at detecting cataracts, but that it may miss certain cases. There is scope for improvement in lowering false positives for glaucoma with 83% accuracy, despite the fact that it displays competitive precision and recall values of 95% and 88% respectively. Finally, the model achieved 91% accuracy, 85% recall, and an F1-score of 88% for the "Normal" class. It is fairly reliable in detecting normal cases, although some normal cases may be categorized as abnormality. The classification results for the proposed ensemble are shown in the confusion matrix. Rows in the matrix correspond to actual classes, whereas columns to predicted classes as shown in Fig. 3.

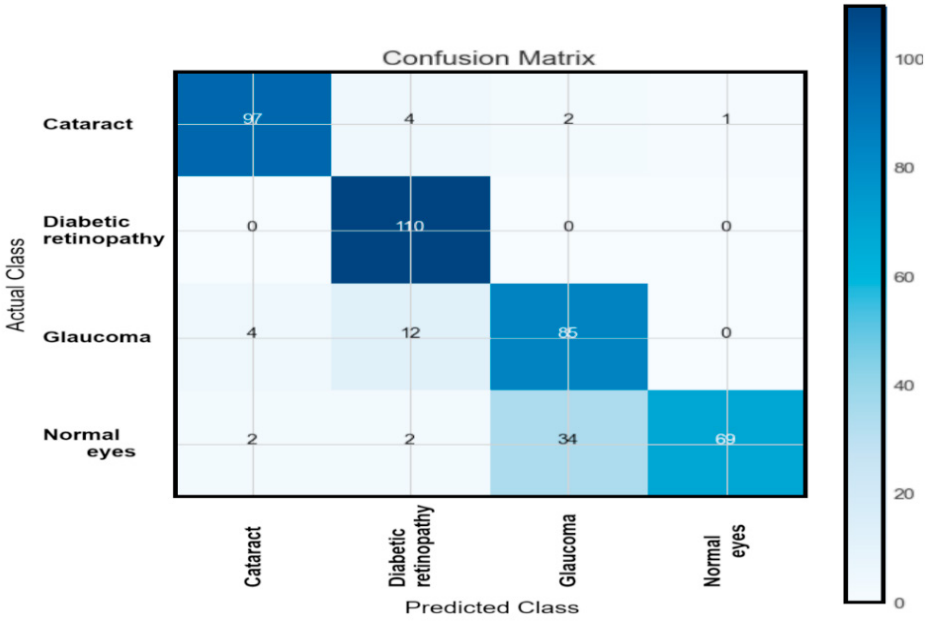


Fig. 3. Confusion matrix illustrating the results of the ensemble model.

The model successfully counts occurrences of diabetic retinopathy with 100% accuracy and recall, resulting in a perfect 110 count. Despite some cases being misclassified as normal eyes and several false positives in the same class, it performs well in identifying glaucoma cases, with 85 genuine positives. It successfully detects 69 occurrences of the "Normal eye" class, although there are false negatives and false positives, giving it an 91% accuracy and 85% recall. Similar errors also occur in the "Cataract" class, which has 89% recall rate, acceptable accuracy of 97%, but occasional false negatives. The matrix emphasizes the model's overall accuracy, notably for cataract and diabetic retinopathy, while indicating areas for refinement in the other classes. The comparison of the proposed ensemble with other state of the art pre-trained models in terms of accuracy is tabulated in Table 3 and pictorially shown in Fig. 4.

Table 3. Comparison of the proposed ensemble with pre-trained models

Disorder class	cataract	diabetic_retinopathy	glaucoma	normal eyes	Overall accuracy
ResNet50	89	98	89	92	90
DenseNet121	94	86	70	99	86
EfficientNetB3	84	95	84	81	87
Proposed	97	100	83	91	93

This analysis highlights the various advantages and enhancements provided by the proposed technique over the alternatives. The model excels in numerous critical areas, including accuracy, recall, f1-score, scalability, distinguishing it as a superior solution for the given situation. This contrast emphasizes the relevance and possible effect of our field study.

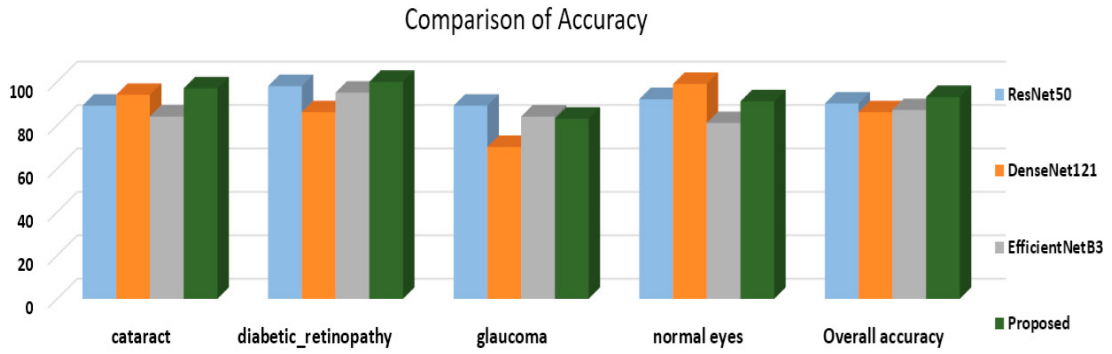


Fig. 4. Comparative analysis of the proposed ensemble with pre-trained models

Thus, the CNN fusion approach proposed in this study is expected to increase diagnostic accuracy in the classification of eye disorders. The model seeks to give more robust and trustworthy predictions by integrating the capabilities of different CNN architectures. Moreover, the fusion of many CNN models is intended to reduce the biases of separate models. This is predicted to yield in a more balanced and impartial approach to eye illness identification, lowering the likelihood of misclassifications and enhancing model fairness overall. The proposed study attempts to increase the generalization of the model to varied datasets by exploiting the collective knowledge retained in multiple CNN architectures via fusion. This is especially critical for the model to function well when dealing with fresh or previously unknown data. Additionally, the CNN fusion method may help the model identify less common eye disorders more effectively by handling unbalanced information more effectively. This solves a prevalent issue in medical imaging when specific illnesses are uncommon.

5. Conclusion and future scope

This study presents a unique fusion strategy for detecting various ophthalmic disorders (viz., like cataract, diabetic retinopathy, glaucoma and normal eyes) using pre-trained models. The integration of different models, like ResNet50, DenseNet 121, and EfficientNetB3, attempts to improve the ophthalmic disorder classification performance overall. The ensemble shows promising results in a thorough examination, attaining a remarkable accuracy of 92% on a broad sample of eye images. With its enhanced diagnostic accuracy, the model may be incorporated into clinical processes as a decision support tool for ophthalmologists. The goal of this implementation is to help healthcare providers make more accurate and fast diagnosis. The model's resilience and generalization capabilities make it acceptable for use in screening programmes, particularly in areas with limited access to specialized eye care. The model might also be linked into telemedicine systems, allowing for remote screening and early diagnosis of eye illnesses. Hence, this method has the potential to have a substantial influence on patient treatment and marks a significant step forward in the incorporation of artificial intelligence into healthcare practices. The capacity of the model to reduce biases and enhance diagnostic accuracy adds to the continued progress of AI-assisted healthcare.

Further research into model combinations and interpretability will surely improve the ensemble's performance and unleash its full potential in increasing eye disease detection and patient treatment. Using transfer learning approaches to fine-tune model combinations based on disease-specific properties might potentially improve accuracy. Investigating interpretability strategies, such as attention processes or feature visualization, would help to increase trust in the ensemble's conclusions and improve its therapeutic value. Furthermore, integration with future technologies such as telemedicine systems has the potential to expand the reach of precise illness diagnosis to remote places. Continuous dataset augmentation and the inclusion of varied patient groups would aid in the resolution of any bias and generalization issues. Collaboration between medical practitioners and machine learning specialists can result in the development of an adaptive and user-friendly diagnostic tool that has a substantial influence on the healthcare system.

Funding

The AICTE funded this project for research promotion under RPS-NER.

References

[1] Balasubramaniam, V. (2021). "Artificial intelligence algorithm with SVM classification using dermascopic images for melanoma diagnosis." J Artif Intell Capsul Netw, 3(1), 34-42.

[2] Sungheetha, A., & Sharma, R. (2021). "Design an early detection and classification for diabetic retinopathy by deep feature extraction based convolution neural network." J Trends Comput Sci Smart Technol (TCSST), 3(02), 81-94.

[3] Gogineni, S., Pimpalshende, A., & Goddumarri, S. (2021). "Eye disease detection using YOLO and ensembled GoogleNet." In Evolutionary computing and mobile sustainable networks (pp. 465-482). Springer.

[4] Abramovich, O., Pizem, H, Eijgen, JV, et al. (2023) "FundusQ-Net: A regression quality assessment deep learning algorithm for fundus images quality grading" Computer Methods and Programs in Biomedicine, 239:107522. doi: 10.1016/j.cmpb.2023.107522.

[5] Tong, Y., Lu, W., Yu, Y., & Shen, Y. (2020). "Application of machine learning in ophthalmic imaging modalities." Eye Vis, 7(1), 1-15.

[6] Chellaswamy, C., Geetha, T. S., Ramasubramanian, B., Abirami, R., Archana, B., & Bharathi, A. D. (2022). "Optimized Convolutional Neural Network based Multiple Eye Disease Detection and Information Sharing System." In Proceedings of the 2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 25–27 May 2022 (pp. 1105–1113).

[7] Khan, M. S. T., Tafshir, N., Alam, K. N. Dhruba, A. R., Khan, M. M., Albraikan, A. A., & Almalki, F. A. (2022). "Deep Learning for Ocular Disease Recognition: An Inner-Class Balance." Computational Intelligence and Neuroscience, 2022, 2022.

[8] Haraburda, P., & Dabała, L. (2022). "Eye Diseases Classification Using Deep Learning." In S. Sclaroff, C. Distante, M. Leo, G. M. Farinella, & F. Tombari (Eds.), Image Analysis and Processing – ICIAP 2022 (pp. 182-195). Springer.

[9] Gour, N., & Khanna, P. (2021). "Multi-class multi-label ophthalmological disease detection using transfer learning based convolutional neural network." Biomedical Signal Processing and Control, 66, 102329.

[10] Dipu, N. M., Shohan, S. A., & Salam, K. (2021). "Ocular Disease Detection Using Advanced Neural Network Based Classification Algorithms." Asian Journal For Convergence In Technology (AJCT), 7(2), 91-99.

[11] Smaida, Mahmoud, and Serhii Yaroshchak (2020). "Bagging of Convolutional Neural Networks for Diagnostic of Eye Diseases." International Conference on Computational Linguistics and Intelligent Systems.

[12] Patil, Y., Shetty, A., Kale, Y., et al. (2023). "Multiple ocular disease detection using novel ensemble models." Multimedia Tools and Applications. Advance online publication. https://doi.org/10.1007/s11042-023-16000-w

[13] Koonce, B. (2021). "ResNet 50." In Convolutional Neural Networks with Swift for TensorFlow (pp. 85-102). Apress.

[14] Batool, A., & Byun, Y.-C. (2023). "Lightweight EfficientNetB3 Model Based on Depthwise Separable Convolutions for Enhancing Classification of Leukemia White Blood Cell Images." IEEE Access, 11, 37203-37215. https://doi.org/10.1109/ACCESS.2023.3266511

[15] Chhabra, M., & Kumar, R. (2022). "A Smart Healthcare System Based on Classifier DenseNet 121 Model to Detect Multiple Diseases." In N. Marriwala, C. Tripathi, S. Jain, & D. Kumar (Eds.), Mobile Radio Communications and 5G Networks (pp. 267-277). Springer.

[16] Porwal, P., Pachade, S., Kamble, R., Kokare, M., Deshmukh, G., et al. (2018). "Indian Diabetic Retinopathy Image Dataset (IDRiD)." IEEE Dataport. https://dx.doi.org/10.21227/H25W98

[17] EyePACS Dataset. http://www.eyepacs.com.

[18] Sadiq, Muhammad Tariq, Aziz, Muhammad Zulkifal, Almogren, Ahmad, Yousaf, Adnan, Siuly, Siuly, Rehman, Ateeq Ur (2022). "Exploiting pretrained CNN models for the development of an EEG-based robust BCI framework." Computers in Biology and Medicine, 143, 105242. https://doi.org/10.1016/j.compbiomed.2022.105242

[19] Singh, M.K., Kumar, B. (2023). "Fine Tuning the Pre-trained Convolutional Neural Network Models for Hyperspectral Image Classification Using Transfer Learning." In P.K. Shukla, K.P. Singh, A.K. Tripathi, & A. Engelbrecht (Eds.), Computer Vision and Robotics: Algorithms for Intelligent Systems (pp. 269-279). Springer.

[20] Khozeimeh, F., Sharifrazi, D., Izadi, N.H., et al. (2021). "Combining a convolutional neural network with autoencoders to predict the survival chance of COVID-19 patients." Scientific Reports, 11, 15343. https://doi.org/10.1038/s41598-021-93543-8

[21] Voon, W., Hum, Y.C., Tee, Y.K., et al. (2022). "Performance analysis of seven Convolutional Neural Networks (CNNs) with transfer learning for Invasive Ductal Carcinoma (IDC) grading in breast histopathological images." Scientific Reports, 12, 19200. https://doi.org/10.1038/s41598-022-21848-3

[22] Gheisari, S., Shariflou, S., Phu, J., et al. (2021). "A combined convolutional and recurrent neural network for enhanced glaucoma detection." Scientific Reports, 11, 1945.

[23] Ocular disease recognition. https://www.kaggle.com/andrewmvd/ocular-disease-recognition-odir5k.

[24] Budai, A., Bock, R., Maier, A., Hornegger, J., Michelson, G. (2013). "Robust Vessel Segmentation in Fundus Images." International Journal of Biomedical Imaging, 2013.

[25] Biswas, A., Banik, R. (2023). "Estimating Risk Levels and Epidemiology of Diabetic Retinopathy using Transfer Learning." In Proceedings of the 2023 Third International Conference on Secure Cyber Computing and Communication (ICSCCC) (pp. 287-292). https://doi.org/10.1109/ICSCCC58608.2023.10176908.

[26] Keerthana, I. P., Kumar, R. S. (2023). "A Survey of Deep Learning Models to Detect and Classify Eye Disorders." In 2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS) (pp. 30-36). https://doi.org/10.1109/ICSCSS57650.2023.10169846.

[27] Hicks, S. A., Strümke, I., Thambawita, V., Hammou, M., Riegler, M. A., Halvorsen, P., Parasa, S. (2022). "On evaluation metrics for medical applications of artificial intelligence." Scientific Reports, 12, 5979. https://doi.org/10.1038/s41598-022-099.